

Introduction to Biophysics II: Quantum physics & biology

Lecture 08: Energy transfer in photosynthesis - III

Förster Resonance Energy Transfer:

- FRET Applications
- Modeling FRET kinetics of complex system

Reading:

Modern Optical Spectroscopy, chapters 4,5, 7

Additional:

Quantum Effects in Biology, chapter 3

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Modeling energy transfer kinetics

Example: a system of 4 interacting pigments (1,2,3,4) where one of the pigments (4) serves as both the reaction center and antenna pigment.

Denote k_{ij} – energy transfer rates between the pigments
 τ_i – excited state lifetime of pigments $i=1,2,3,4$
 $N_i(t)$ – time-dependent population of excited pigments
 k_{45} – charge separation time (assume irreversible)

Can write down a system of kinetic equations:

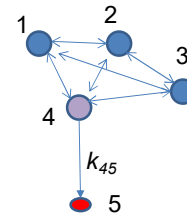
$$\frac{dN_1}{dt} = -N_1 \left(k_{12} + k_{13} + k_{14} + \frac{1}{\tau_1} \right) + N_2 k_{21} + N_3 k_{31} + N_4 k_{41}$$

$$\frac{dN_2}{dt} = -N_2 \left(k_{21} + k_{23} + k_{24} + \frac{1}{\tau_2} \right) + N_1 k_{12} + N_3 k_{32} + N_4 k_{42}$$

$$\frac{dN_3}{dt} = \dots$$

$$\frac{dN_4}{dt} = -N_4 \left(k_{41} + k_{42} + k_{43} + k_{45} + \frac{1}{\tau_4} \right) + N_1 k_{14} + N_2 k_{24} + N_3 k_{34}$$

$$\frac{dN_5}{dt} = N_4 k_{45}$$

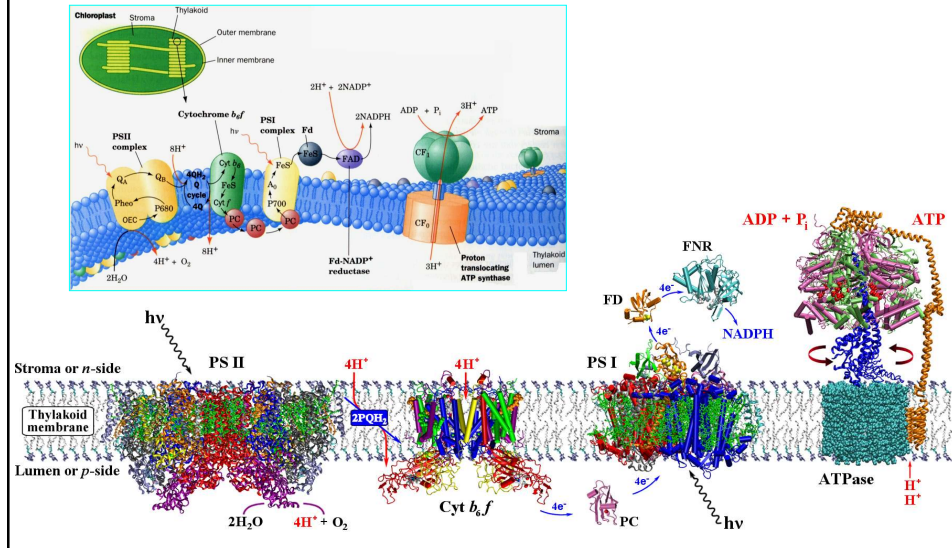


Note: assumed fast transfer, ignored radiative and nonradiative decay channels
 (can be included as, for example, for first equation adding term $-N_1 k_1$, with k_1 rate of decay)

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FRET of complex system: PS I

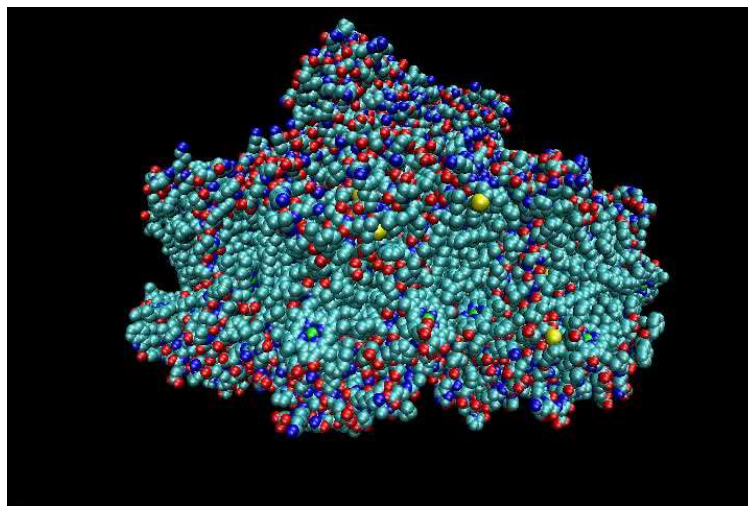
Example: photosystem I



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FRET in complex system: PS I

Example: photosystem I



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Analyzing FRET kinetics data

Example: photosystem I: 96 pigments, 96x96 energy transfer rates

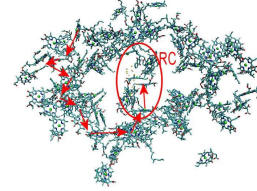
Solve System of 96 rate equations:

$$\begin{cases} \frac{dn_1}{dt} = -\left(\frac{1}{\tau_{1f}} + k_{21} + k_{31} + \dots + k_{n1}\right)n_1 + k_{12}n_2 + k_{13}n_3 + \dots + k_{1n}n_n \\ \frac{dn_2}{dt} = -\left(\frac{1}{\tau_{2f}} + k_{12} + k_{32} + \dots + k_{n2}\right)n_2 + k_{21}n_1 + k_{23}n_3 + \dots + k_{2n}n_n \\ \dots \end{cases}$$

In principle can be solved analytically, but the resulting equation will be much too long...

Solve on computer:

1. Set initially excited molecule (k) population to N_k to 1
Leave all others $N_{i \neq k} = 0$. Time $t=0$
2. Increase time by a small amount dt . Use the equations above to find changes in population dN_i after time dt
3. Repeat step 2 until needed to generate $N_i(t)$ profiles
4. Assuming spectra for each excited state are known – generate $S(\lambda, t)$ and compare with the experiment



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Analyzing FRET kinetics data

Example: photosystem I: 96 pigments, 96x96 energy transfer rates

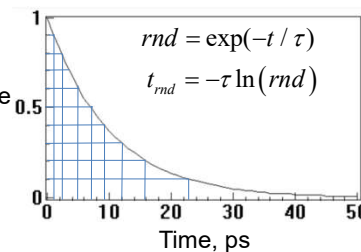
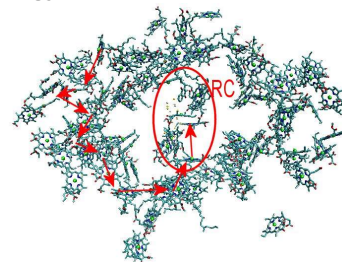
Alternative: **Monte Carlo**

1. Define which molecule gets excited.
Can be based on the absorption spectra of each pigment and defined photon wavelength
2. Weighted on k_{ij} generate "random" transfer time τ_{ij}
to every other molecule in the system.
3. Pick the shortest τ_{ij} , record that the excitation "hopped" to that pigment j at time $t + \tau_{ij}$
4. Go to step 2 to see where and when the excitation "hops" next

What you need: a random number generator that produces an exponential distribution of events in time

How can one do that?

Generate random number rnd between (0,1)
Project to exponent, find respective time t_{rnd}



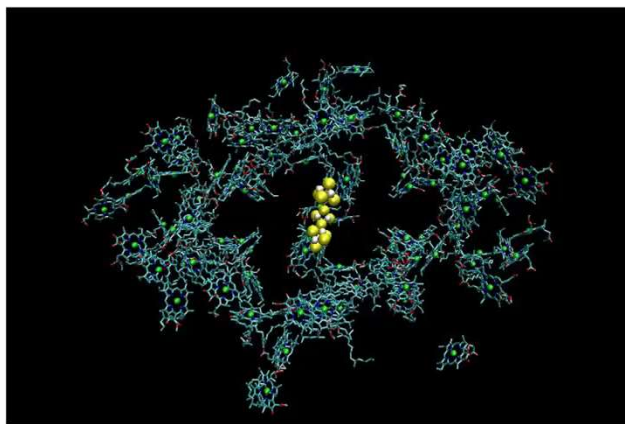
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Analyzing FRET kinetics data

Example: photosystem I: 96 pigments, 96x96 energy transfer rates

Alternative: **Monte Carlo**

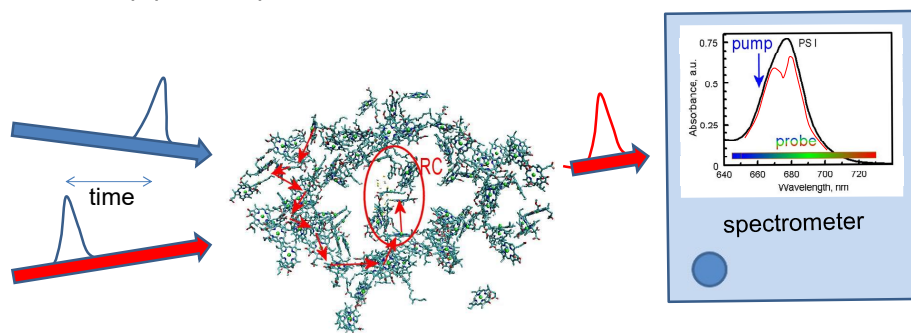
Time slowed down a million million times! $1\text{ps} = 10^{-12}\text{s} \rightarrow 1\text{s}$



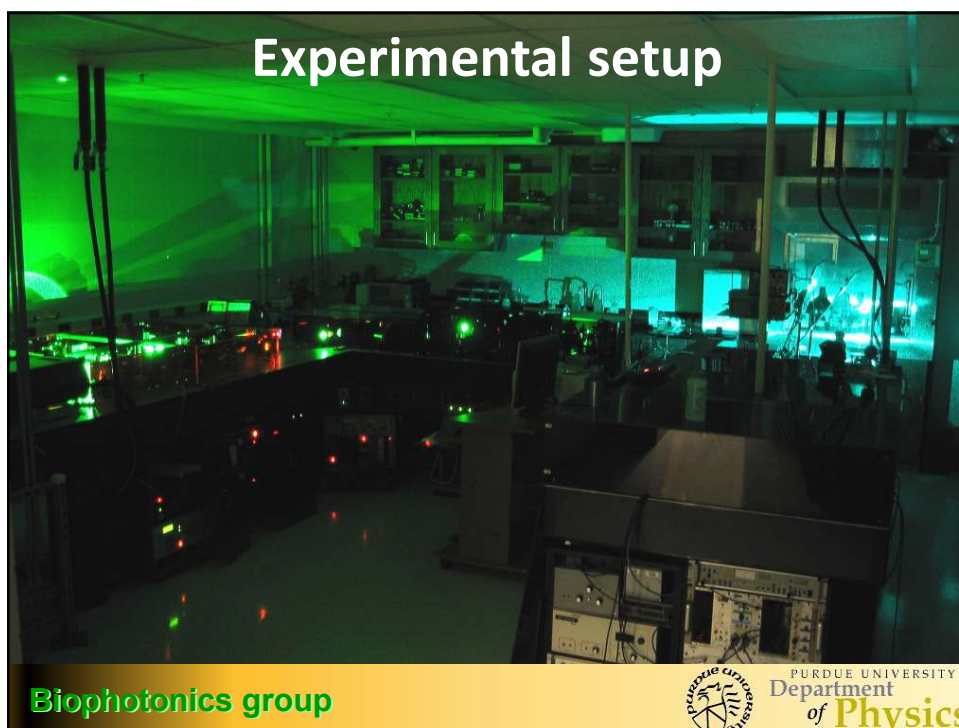
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FRET in complex system: PS I

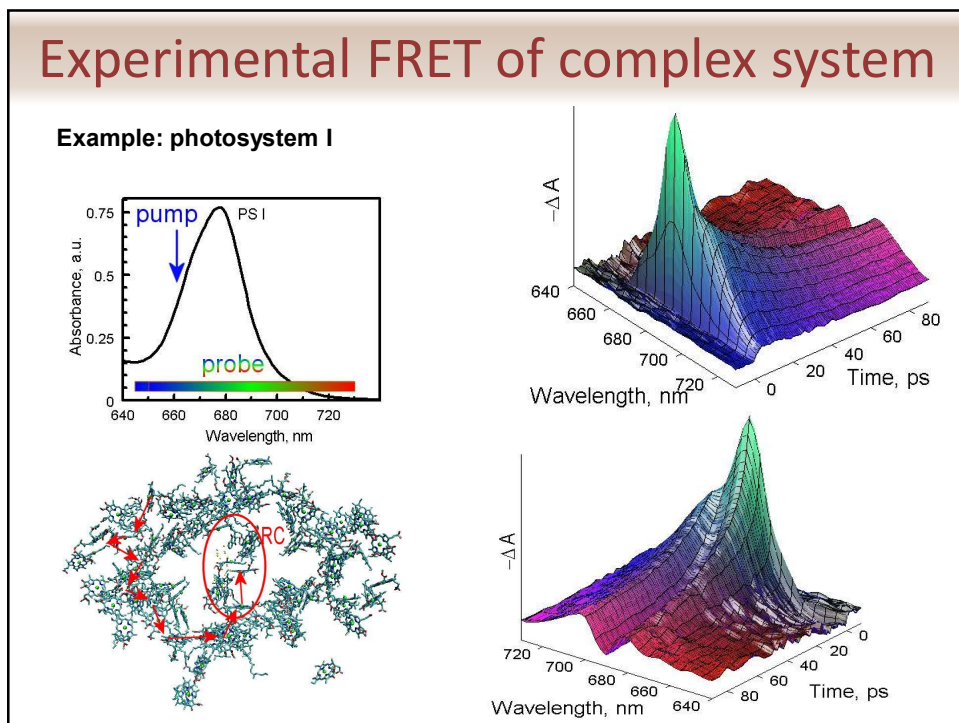
Pump-probe experiment



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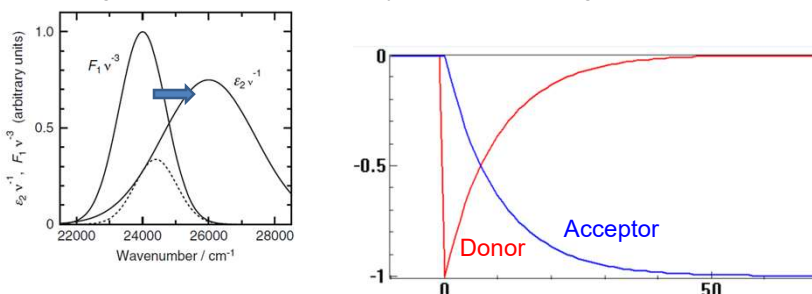
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Analyzing FRET kinetics data

Using pump-probe spectroscopy: absorption changes



Note: analysis is more complicated than fluorescence.

signal is comprised of 3 parts:

PB – photobleaching, the number of absorbing molecules becomes smaller

SE – stimulated emission: excited molecules can *amplify* light (Einstein)

ESA – excited state absorption: excited molecules can absorb light going to higher excited states

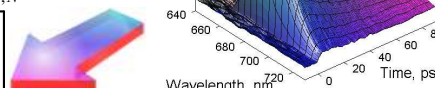
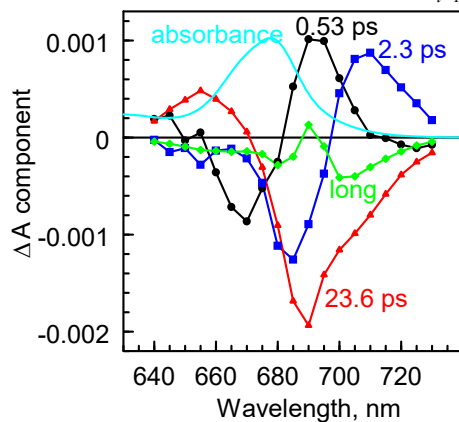
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Analyzing FRET kinetics data

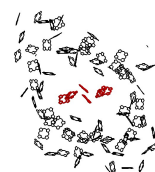
Example: photosystem I

Decay associated spectra (DAS)

$$S(\lambda, t) = \sum_{i=1, N} A_i(\lambda) e^{-t/\tau_i}$$



0.53 ps - fast equilibration
 2.3 ps - slow equilibration
 23.6 ps - antenna → RC
 Long - P700⁺



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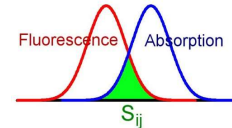
Pump-probe spectroscopy of PS I

What can we learn from the theory and the experiment?

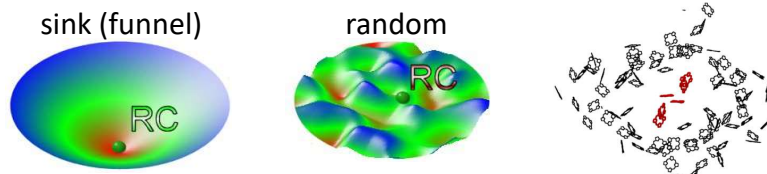
96 pigments: $96 \times 95 = 9120$ rate constants

$$k_{ij} = S_{ij} \frac{\left[\hat{\mu}_i \cdot \hat{\mu}_j - 3(\hat{\mu}_i \cdot \hat{R}_{ij})(\hat{\mu}_j \cdot \hat{R}_{ij}) \right]^2}{R_{ij}^6}$$

Hard to compute S_{ij} of the Chls



It would be thermodynamically advantageous to position lower energy Chls closer to the reaction center to get faster and unidirectional transfer toward it (sink model).



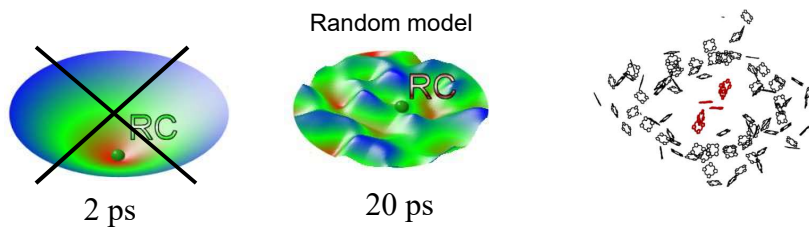
Comparison of simulations with experiment supports *random* model

The energy transfer is still fast enough to be almost 100% efficient.

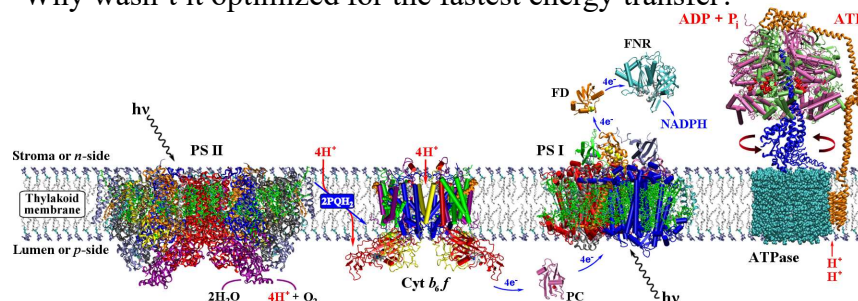
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Pump-probe spectroscopy of PS I

What can we learn from the theory and the experiment?



Why wasn't it optimized for the fastest energy transfer?



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FRET application: measuring distance

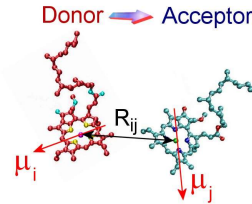
Application:
use fluorescence yield as a measure of distance
between donor and acceptor

Measure fluorescence intensity of donor in the absence (Φ) and in the presence (Φ_q) of acceptor:

$$\frac{\Phi}{\Phi_q} = \frac{k_r/k_{tot}}{k_r/(k_{tot} + k_{rt})} = 1 + \frac{k_{rt}}{k_{tot}} = 1 + k_{rt}\tau$$

$$|R_{21}| = R_o(k_{rt}\tau)^{-1/6} = R_o(\Phi/\Phi_q - 1)^{-1/6}$$

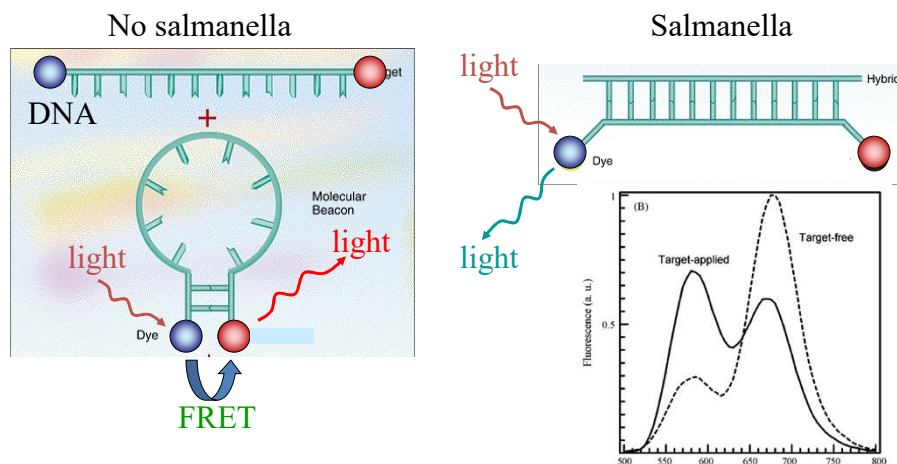
Alternatively: measure decay of donor excited state directly with and without acceptor



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FRET: application

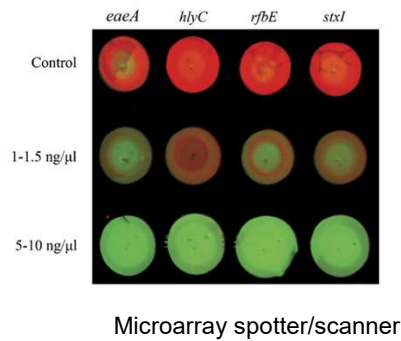
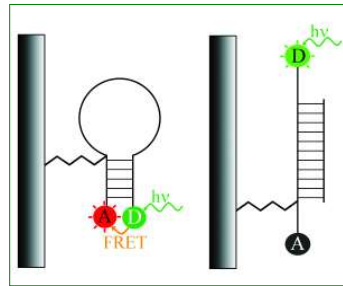
Example: food pathogen detection using DNA “beacon”
fluorescence FRET markers: CY-3 & CY-5



Kim et al., Biosensors and Bioelectronics, 22, p.1041 (2007)

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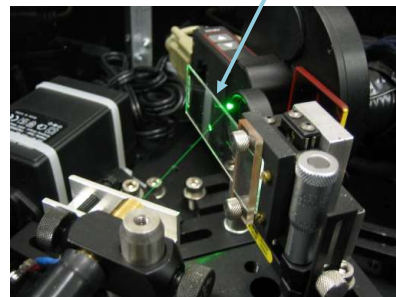
FRET: application



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FRET: application

Example: food pathogen detection using DNA “beacon”
fluorescence FRET markers: CY-3 & CY-5

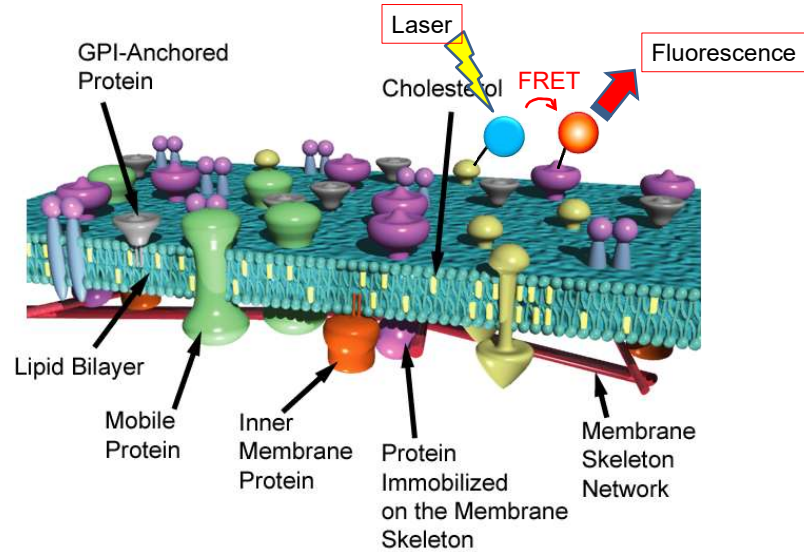


Kim et al., Biosensors and Bioelectronics,
22, p.1041 (2007)

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FRET: application

FRET markers: studying protein-protein interactions

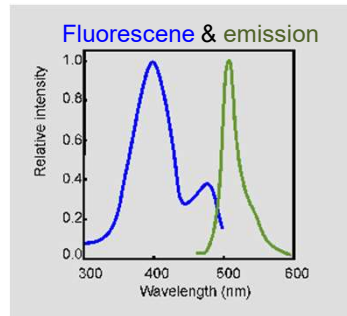
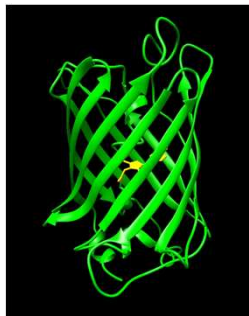
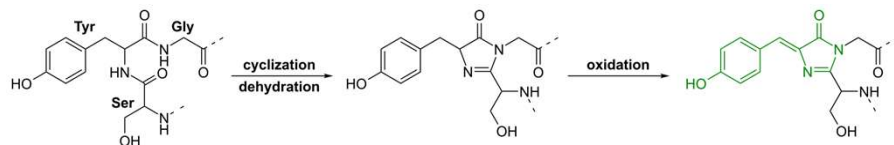


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FRET: application

Dye-free markers: green fluorescent protein (GFP)

protein first isolated from the [jellyfish *Aequorea victoria*](#)



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FRET: application

Dye-free markers: green fluorescent protein (GFP): can incorporate into other proteins attaching respective DNA sequence



Mice with GFP inserted into their genomes for neurology studies

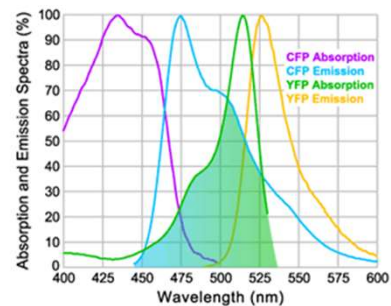
(No FRET, just looks cool....)

https://proteopedia.org/wiki/index.php/Green_Fluorescent_Protein

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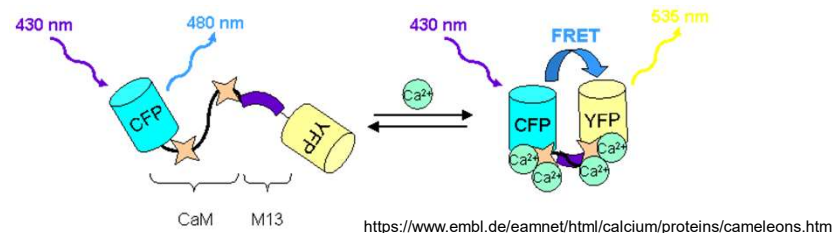
FRET: application

Variants of GFP: different spectra
Can incorporate DNA sequence of GFP-like protein fragments into proteins of interest.
For example: CFP (cyan) and YFP (yellow)



<https://www.semrock.com/fret.aspx>

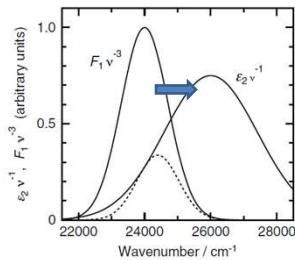
Example: "cameleon": calcium detector



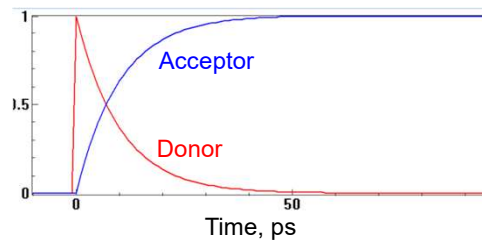
<https://www.embl.de/eamnet/html/calcium/proteins/cameleons.htm>

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FRET: Time-resolved measurements



Using Fluorescence:



Most direct way of measurement, works well for even very fast FRET

Compare to static measurements using changing donor fluorescence yield:

$$\frac{\Phi}{\Phi_q} = 1 + k_{rt}\tau \quad k_{rt} = \left(\frac{\Phi}{\Phi_q} - 1 \right) / \tau$$

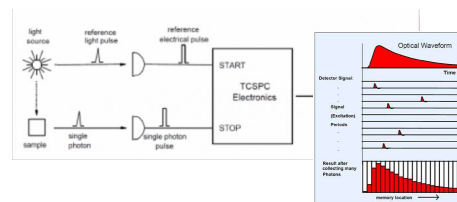
Fast transfer rate $\rightarrow$ small $\Phi_q \rightarrow$ hard to measure as donor and acceptor fluorescence spectra often overlap

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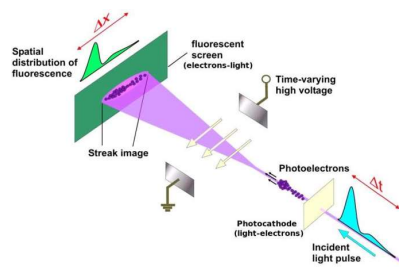
FRET: Time-resolved measurements

Fluorescence lifetime measurements

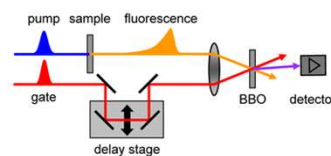
1. Time-correlated photon counting (>10ps)



2. Streak camera (~1 ps)



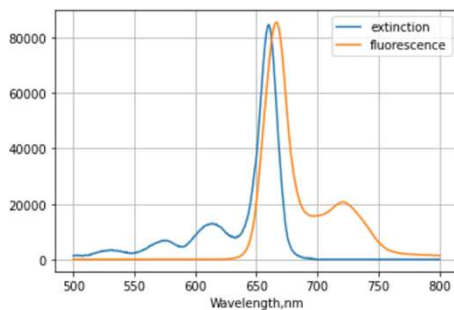
3. Fluorescence up-conversion (<0.1 ps)



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Homework 2

Problem 02. Förster Radius as a function of spectral shift between Chl a molecules



Use Lecture 8:

$$R_o = 9.80 \times 10^3 (\kappa^2 \Phi_n^{-4} \bar{J})^{1/6} \text{ \AA}$$

$$\bar{J} = \frac{\int F_1(\bar{\nu}) \epsilon_2(\bar{\nu}) \bar{\nu}^{-4} d\bar{\nu}}{\int F_1(\bar{\nu}) d\bar{\nu}} = \frac{\int F_1(\lambda) \epsilon_2(\lambda) \lambda^2 d\lambda}{\int F_1(\lambda) \lambda^{-2} d\lambda}$$

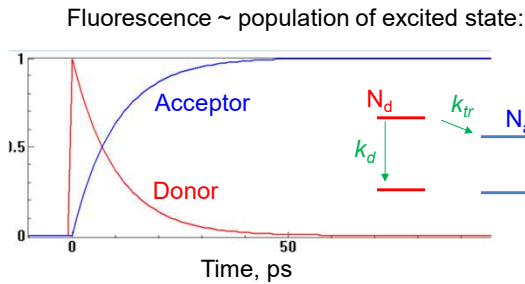
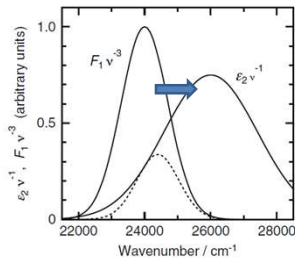
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EXTRA

Skip the following slides...

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Analyzing FRET kinetics data



System of kinetic equations: (assume unidirectional transfer)

$$\begin{aligned} \frac{dN_d}{dt} &= -k_d N_d - k_{tr} N_d = -(k_d + k_{tr}) N_d & N_d &= N_0 \exp(-(k_d + k_{tr})t) \\ \frac{dN_a}{dt} &= -k_a N_a + k_{tr} N_d & N_a &= N_0 \{ \exp(-k_a t) - \exp(-(k_d + k_{tr})t) \} \end{aligned}$$

N_d , N_a – populations of excited donors and acceptors

k_d – lifetimes of excited donor and acceptor in absence of energy transfer

k_{tr} – energy transfer rate

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Analyzing FRET kinetics data

$$\begin{aligned} N_d &= N_0 \exp(-(k_d + k_{tr})t) \\ N_a &= N_0 \{ \exp(-k_a t) - \exp(-(k_d + k_{tr})t) \} \end{aligned}$$

Spectral shapes:

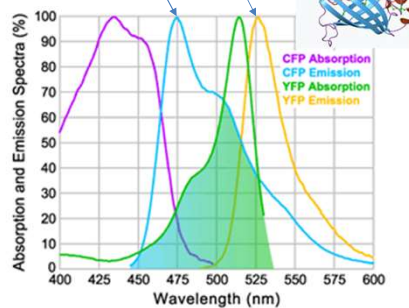
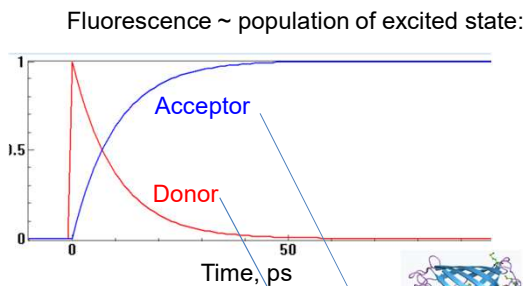
F_a and F_d

Fluorescence spectrum of donor is superseded by fluorescence spectrum of acceptor as energy is transferred.

Fluorescence spectrum kinetics:

$$\begin{aligned} F(\lambda, t) &= F_d(\lambda, t) + F_a(\lambda, t) \\ F(\lambda, t) &= F_d(\lambda) \exp(-(k_d + k_{tr})t) + \\ &+ F_a(\lambda) \{ \exp(-k_a t) - \exp(-(k_d + k_{tr})t) \} \end{aligned}$$

Limited number of lifetimes should fit kinetics at every wavelength!



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Analyzing FRET kinetics data

Global fit:

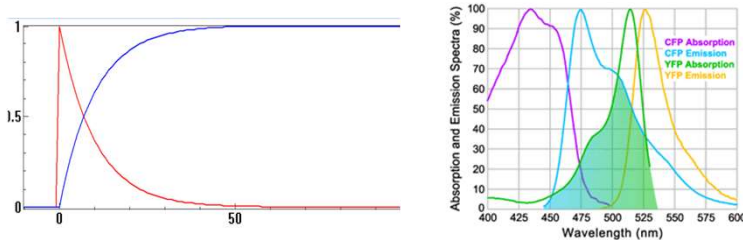
Fit set of kinetic time-dependent signal profiles (fluorescence or absorption) at every wavelength with some limited (minimal) number of rate constants (decay times):

$$S(\lambda, t) = \sum_{i=1, N} A_i(\lambda) e^{-k_i t}$$

Each lifetime will be associated with spectrum, DAS

DAS – Decay Associated Spectrum

Each lifetime/rate will have its unique spectrum which facilitates interpretation of data



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Note on random number generator with arbitrary distribution

$D(r)$ - Desired random number distribution

$I(r)$ – integral of $D(r)$ normalized to 1

$$I(r) = \frac{\int_{-\infty}^r D(r) dr}{\int_{-\infty}^{\infty} D(r) dr}$$

Define inverse function $I'(x)$:

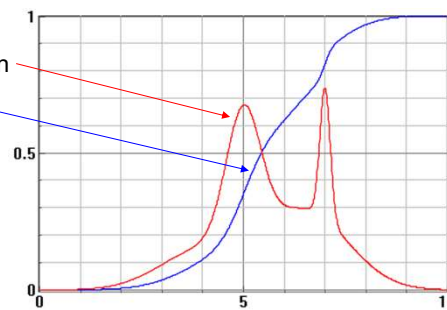
$$I(r) = x$$

$$\bar{I}(x) = r$$

Random number r with distribution $D(r)$ can be now found from a standard random number x generated between (0,1):

$$r = \bar{I}(x)$$

Why didn't we integrate for exponential distribution?



Distribution can be a measured curve

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